

Generative AI Mini Project

AI Agent for Research Assistance

Objective

Build a LangGraph-powered AI agent that can:

- Accept a research question
- Decide which tool to use
- Gather information from relevant sources
- Generate a clear and structured response

LLM Setup

Choose one of the free LLM options.

Part 1: Build Tools

Create and test the following tools:

1. Wikipedia Tool

Input: Topic

Output: Summary from Wikipedia

2. Arxiv Tool

Input: Research topic

Output: Top 3 paper abstracts

3. LLM Knowledge Tool

Input: Query

Output: Direct response using the LLM

Part 2: Build the AI Agent (LangGraph)

Design an agent with the following components:

- Router Node – selects the appropriate tool
- Tool Node – executes the selected tool
- Synthesiser Node – generates the final response

Requirements

- Define the agent state
- Connect nodes using LangGraph

- Implement conditional routing for out-of-scope queries

Part 3: Testing

Run the agent on the following queries:

- What is Retrieval-Augmented Generation?
- What are the latest research papers on large language models?
- Who invented the Transformer architecture?
- What is the capital of Australia?
- One question of your choice

Output Requirements

For each query, include:

- Tool selected
- Final response

Part 4: Conversation Memory

Extend the agent to support multi-turn conversations.

Test Conversation

- What is a Transformer?
- How is it different from an RNN?
- Which one would you use for text generation and why?

Part 5: Evaluation

Evaluate responses based on:

- Accuracy
- Relevance
- Groundedness

Bonus (Optional)

Build a Streamlit UI with:

- Chat input
- Response display
- Tool usage visibility
- Conversation history

Deliverables

Choose one:

Option A: Jupyter Notebook

Complete implementation with outputs

Option B: Streamlit Application

Working application